

DR. JAEHYUK LEE, PH.D.

☎ 470-861-6020 🏠 <https://ruach.github.io> ✉ jaehyukl@nvidia.com

SUMMARY

I work on high-assurance systems security: Trusted Execution Environments, cryptographic attestation and root-of-trust, and hypervisor/OS isolation, paired with the adversarial side (side-channel attacks, exploit development, and fuzzing) to test whether those guarantees hold. Throughout my career, I have tackled security problems across the full hardware-software stack, driven by a desire to turn complex research problems into production-ready solutions.

TECHNICAL SKILLS

Software: C, C++, Rust, Scala, Python, TF-A, TF-RMM, Caliptra FW, PSC FW (NVIDIA IROT), Linux Kernel, KVM, QEMU, ARM/X86 Assembly, Fuzzing, GDB, CoCLI
Hardware: GEM5 (x86 simulator), Processor Pipeline, Cache, TLB, Microcode, HW Side Channel, (System)Verilog, Chisel (RISC-V), Intel SGX & TDX, ARM CCA & TrustZone, IOMMU, PCIe
Spec: Entity Attestation Token (EAT), Concise Reference Integrity Manifest (CoRIM), DICE Protection Environment (DPE), Security Protocol and Data Model (SPDM), TEE Device Interface Security Protocol (TDISP)

WORK EXPERIENCE

Senior Security Engineer: NVIDIA Corporation, Santa Clara, CA Jul 2024 – Current

- Owned an end-to-end cryptographic attestation chain from silicon root of trust to signed evidence for Vera and Grace CC
 - Developed cross-layer firmware and software stacks across bootloader, RMM, ATF, PSCFW, and Caliptra DPE to enable end-to-end attestation and evidence generation for Vera CCA and Baremetal Secure AI on Vera and Grace CPUs.
 - Developed and deployed CoRIM/CoMID manifests for Vera CCA and Vera/Grace baremetal attestation
- Led Live Firmware Activation (LFA) support for RMM, building the software stack across *RMM, ATF, and Vera IROT*.
- Hardened guest OS boot sequence and model loading using initramfs disk encryption with early, attested key provisioning.

Postdoc Researcher: Georgia Institute of Technology, Atlanta, GA Jan 2022 – Jul 2024

- Developed an automated fuzzer selection tool that dynamically selects fuzzer(s) based on runtime analysis [3]
- Designed and implemented a hardware-enforced secure I/O path between confidential VMs and DMA peripherals [1]
- Conducted vulnerability research on Intel and ARM's latest hardware-isolated confidential VM architectures (TDX & CCA)

Postdoc Researcher: Korea Advanced Institute of Science and Technology, Korea Mar 2021 – Dec 2021

- Demonstrated an eviction-free cache side-channel attack on x86 and Apple M1, bypassing eviction-monitoring defenses [8]
- Took the lead in managing and mentoring a group of 20 Ph.D. students

Research Intern: Georgia Institute of Technology, Atlanta, GA Nov 2019 – May 2020

- Developed a hardware-based side-channel mitigation system on GEM5, monitoring cache and TLB eviction events [2]

Research Intern: Microsoft Research, Redmond, WA May 2016 – August 2016

- Executed practical ROP attacks on Intel SGX by exploiting memory corruption vulnerabilities without binary details [4]
- Developed a secure CDN that enables enterprises to offload sensitive user data to untrusted third-party cloud providers

PUBLICATION & PATENT (828 citations)

top-tier conferences (SP, USENIX Security, NDSS) and Journal (TDSC)

- PORTAL: Fast and Secure Device Access with Arm CCA for Modern Arm Mobile SoC** | [Paper](#) SP 2025
 - Eliminated per-transfer encryption overhead for Arm CCA Realm VMs accessing DMA-capable peripherals.
 - Extended the Linux SMMU driver to delegate page table management to RMM; implemented the Realm-side driver and the full RMI call path across KVM, TF-RMM, and the SMMU hardware controller. Changed RMM implementation to enable hardware-enforced (GPT) isolated DMA regions without cryptographic operation.
- SENSE: Enhancing Microarchitectural Awareness via Subscription-Based Notification** | [Paper](#) NDSS 2024
 - Addressed the lack of microarchitectural visibility in TEEs that leaves them blind to cache/TLB side-channel attacks
 - Extended the GEM5 out-of-order CPU pipeline with new micro-ops and added event-tracking metadata bits to the cache and TLB models; implemented the enclave-side handler dispatch path.
- autofz: Automated Fuzzer Composition at Runtime** | [Paper](#) | [Hacker News](#) USENIX 2023
 - Eliminated the need for per-target fuzzer selection by building a runtime coordinator that measures each fuzzer's coverage
 - Implemented the orchestration layer in Python using Linux cgroups for CPU pinning; integrated 12 fuzzers (AFL, AFLFast, MOpt, RedQueen, QSYM, Angora, and others) under a unified coverage-feedback interface.
- Hacking in Darkness: Return-oriented Programming against Secure Enclaves** | [Paper](#) | [Talk](#) | [Demo](#) USENIX 2017
 - Demonstrated the first ROP attack against Intel SGX: exploited the deterministic AEX fault oracle to probe enclave memory word-by-word and reconstruct a gadget map *without any binary knowledge* and showcased full key-exfiltration

- Built the AEX-based memory scanner and ROP-chain generator in C/x86 assembly; reverse-engineered Intel SGX library (e.g., EPCM/TCS/SSA layouts) to locate gadgets and craft the exploit entirely from outside the enclave.
- [5] **SGX-Bomb: Locking Down the Processor via Rowhammer Attack** | [Paper](#) | [Hacker News](#) | [Demo](#) **SysTEX 2017**
- Showed that an enclave can weaponize Rowhammer to permanently lock the Memory Encryption Engine (MEE) and force a machine reboot — rendering SGX’s integrity guarantees a denial-of-service vector for cloud multi-tenant scenarios.
 - Wrote a kernel driver to reverse-engineer DRAM bank/row mapping from physical addresses; implemented the double-sided Rowhammer loop in x86 assembly inside an enclave to flip MEE metadata bits and trigger a machine lockdown.
- [6] **PrivateZone: Providing a Private Execution Environment using ARM TrustZone** | [Paper](#) **TDSC 2016**
- Extended ARM TrustZone to protect arbitrary user-space processes without porting them to the Secure World, by intercepting Linux scheduling events in Monitor mode and switching per-process stage-2 page tables on each context switch.
 - Implemented the stage-2 page table allocator and the Monitor-mode context-switch handler in ARM assembly; modified the Linux scheduler to emit SVC calls that trigger the switch between trusted and untrusted page tables.
- [7] **OpenSGX: An Open Platform for SGX Research** | [Paper](#) | [Wikipedia: SGX](#) **NDSS 2015**
- Lowered the barrier to SGX research before hardware was available by building an instruction-level emulator in QEMU that faithfully reproduced EPCM, SECS, TCS, and SSA data structures, enabling enclave development and testing on commodity machines.
 - Implemented EENTER/EEXIT context-switch stubs in x86 assembly, EPCM management in QEMU’s softmmu, and a `sgxlib` user library with a stub-call layer that routes enclave system calls through an untrusted trampoline.
- [8] **PRIME+RETOUCH: When Cache is Locked and Leaked** | [Paper](#) **Arxiv**
- Identified a blind spot in eviction-based cache side-channel defenses: Tree-PLRU replacement metadata can be manipulated to “retouch” a cache set so the victim line survives eviction silently — bypassing Intel TSX-based monitors and Apple M1 PMU counters.
 - Reverse-engineered the Tree-PLRU state machine on recent Intel and Apple M1 caches; built the PRIME+RETOUCH PoC in x86/ARM assembly using pointer-chasing and memory barriers; demonstrated cross-core and cross-process secret recovery.
- [9] **Survey On DeFi Users’ Perception of Security Breaches and Countermeasures** | [Paper](#) **USENIX 2024**
- Revealed that DeFi victims overwhelmingly skip post-incident security review and instead seek new platforms to recover losses quickly, suggesting industry regulation is more effective than user education alone.
 - Conducted a two-phase study (14 semi-structured interviews + 493 survey); performed qualitative coding of interview transcripts using Dedoose.
- [10] **A Rule Extraction Method Using Relevance Factor for FMM Neural Networks** | [Paper](#) **KTSDE 2013**
- Enhanced Fuzzy Min-Max (FMM) neural network by incorporating the frequency of features in training.
 - Resolved ambiguity by prioritizing clarity and relevance, while ensuring distinctiveness between features and patterns.
- [11] **Apparatus and method for open and private IoT gateway using Intel SGX** | [Patent](#) **KR102162018B1**
- The invention introduces a secure IoT gateway design, utilizing Intel SGX for protocol conversion and encrypted data transfer. It includes enclave designs for cross-platform communication, encrypted key management, and secure updates.

SOFTWARE ARTIFACTS

Autofz:	https://github.com/sslabs-gatech/autofz	🔗12	★85
OpenSGX:	https://github.com/sslabs-gatech/opensgx	🔗82	★308
SGXBomb:	https://github.com/sslabs-gatech/sgx-bomb	🔗4	★18

EDUCATION

Institution	Degree	Field of Study	Dates	GPA
Korea Advanced Institute of Science and Technology	<i>Ph.D.</i>	Information Security	Aug 2015 - Feb 2021	3.95
Korea Advanced Institute of Science and Technology	<i>M.S.</i>	Information Security	Sep 2013 - Aug 2015	4.1
Handong Global University	<i>B.S.</i>	Computer Science	Mar 2010 - Aug 2013	4.2

PROFESSIONAL ACTIVITIES

- [1] **Journal Review**
- IEEE Transactions on Dependable and Secure Computing [Impact Factor: 7.3, 2024](#)
 - Computer & Security [Impact Factor: 5.7, 2024](#)
 - Journal of Information Security and Applications [Impact Factor: 5.6, 2024](#)
 - IEEE Access [Impact Factor: 3.9, 2024](#)
 - ACM Transactions on Internet Technology [Impact Factor: 3.9, 2024](#)
 - IEEE Journal of Communications and Networks [Impact Factor: 3.6, 2024](#)
- [2] **Conference Review**
- IEEE Mobile Ad Hoc and Smart Systems (MASS) 2024

[3] **External Reviewer**

- IEEE Symposium on Security and Privacy (Oakland)
 - USENIX Security Symposium (Security)
 - ACM Conference on Computer and Communications Security (CCS)
 - Network and Distributed System Security Symposium (NDSS)
 - ACM ASIA Conference on Computer and Communications Security (ASIACCS)
 - ACM Symposium on Operating Systems Principles (SOSP)
 - ACM International World Wide Web Conference (WWW)
- 2016, 2017, 2018, 2020, 2021, 2022
2015, 2021, 2022, 2023, 2024
2015, 2017, 2018, 2019, 2023
2019, 2020, 2021
2015, 2016, 2018
2021
2018

INVITED TALK

[1] **A Novel Cache Side Channel Attack without Attacker Eviction**

- Presented at Intel Side Channel Academic Program (SCAP), Virtual, Sep 2020

[2] **Hacking in Darkness: Return-oriented Programming against Secure Enclaves**

- Presented at the 26th Usenix Security | [Presentation Link](#), Vancouver, BC, Canada, Aug 2017
- Presented at KimchiCon | [Presentation Link](#), Daejeon, South Korea, Aug 2017

[3] **Tutorial: Trusted Execution Environment: TrustZone, ITP and SGX**

- Korea Institute of Information Security and Cryptology, Seoul, South Korea, Apr 2015

THESIS

<i>Unintended Consequences of System Design: Unpremeditated Usage of Benign System Components Devastates Security</i>	Ph.D. Thesis.
<i>Detecting Emulated Environment: Exploiting Behavioral Discrepancies In QEMU</i>	M.S. Thesis.